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New ways of thinking design pedagogy: body-based methods in design education

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ABSTRACT: Designing differently to cope with ever-changing world challenges and uncertainty requires a pedagogical shift for new ways of thinking to be taught, practiced and trained within the educational programs. These can allow teachers and students to question not only how designers solve problems but how designers identify problems, and the assumptions and epistemologies associated with this. Using body-based and movement-based methods in a design course, we practiced different kinds of attentive modes in site mapping to entice a sense of discovery, learning and reflection. We encouraged students to ask why and how such modes of attention might produce different or 'better' design outcomes, and how attending differently could offer different perspectives on their own role as designers within society. Equipping designers in such a way is essential to enable them to question and think beyond current ways of devising solutions and can potentially create awareness of the implications of their own practices.

1 INTRODUCTION AND THEORETICAL CONTEXT

Collaborating as an urban design and an arts practice researcher, we have been investigating how to bring about the conditions for new ways of thinking through a multi-disciplinary approach in the design education context. In an urban design course module of the architecture and design program of Aalborg University, we introduced artistic and body-based tools and assignments to enable students to discover and explore a much wider range of modes of attention to spatial experience than conventional design methods tend to offer.

In Aalborg University (AAU), educational activities are framed under the Problem-Based-Learning (PBL) pedagogical model. PBL in AAU is characterized by project work based on authentic problems (real world problems), self-governed group work and collaboration with organizations and external stakeholders (Askehave et al 2015). Problem-based project work provides opportunities for independent learning since problems are defined by students based on their own motivations and curiosity and the interactions with external parties. Project outcomes should be exemplary, meaning that the results and learnings should be transferable to other professional situations. Students are guided by theoretical and analytical frameworks under the supervision of one or two supervisors (Servant-Miklos 2020). This pedagogical approach is student-centered and facilitates experiential learning: learning by doing (with others).

Experiential learning is based on Kolb's model and theory of reflecting learning which explains that students learn better when they experiment (learning by doing things) thereby constructing their own knowledge that can be subsequently re-applied or implemented in new and similar situations (Kolb 1976 in Bindal 2022, 93). According to Bindal (2022, 93) experiential learning forms the basis for problem-based learning in design studio educations 'as

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a process of experimenting and forming one's own experience and, henceforth, reflecting on those experiences and forming new abstract concepts.'

However, according to Savin-Baden (2020, 4), PBL pedagogies are hampered by institutional narratives that privilege professional development, the acquisition of technical skills and knowledge applicability over life-long learning (critical reflection skill development and knowledge transformation). In addition, recent literature in architectural pedagogy also argues for an evolving shift of the teaching pedagogies and even professional practices. For instance, recent world circumstances, specifically the pandemic Covid-19 and climate change, have shown that world challenges are becoming more complex: uncertain and unpredictable (Sanderson and Stone, 2022). To cope with it the (design) profession is in flux (Cairns in Sanderson and Stone, 2022, p. xvi).

Following the above-mentioned critiques, Bermudez (2022, 17) argues that higher education has become 'superficial, secular and utilitarian. Universities and colleges have continued to apply their familiar cognicentric-instrumental paradigm because of its more efficient capacity to market, package and measure educational programs, curricula, and outcomes respectively. . .outdated discipline-narrow teaching methods is not how to prepare students for our complex 21st Century.' Bermudez proposes that architectural education and pedagogy in universities should assist and support students in their spiritual development – meaning in their process of self-discovery, meaning-making and personal transformation. This is assisting them to answer 'big questions' such as '… *"Who am I?" "What are my most deeply felt values?" …"What sort of world do I want to help create?" ….*' to help them to become better human beings (Astin, Astin and Lindholm in Bermudez 2022, 17, emphasis in original). He contends that this process of self-discovery (subjective-oriented) should complement the object of design inquiry (mediating big questions with design questions) and this balance should be found in the interplay between curriculum, course content and pedagogy (2022, 27).

Skrzypczak (2021) connects Bermudez' call for shifting the focus of design education towards questioning of values and purpose, with the adoption of somatic methods – that is, methods of spatial design that foreground and take seriously knowledge of, and arising within, the body. He notes that introducing somatic movement methods in architectural practice could effect 'a shift in architects' value systems' (2021, 59). Skrzypczak explains that although the emphasis on a first-person perspective in somatic or body-based methods is evidently connected to phenomenology, "the dominant mode of inquiry in phenomenology is thinking, in somatics it is moving" (2021, 54).

Bingham-Hall & Cosgrave (2019) agree that traditional approaches in engineering favour a one-way approach which focuses on solving problems rather than questioning how problems are identified; they suggest that adopting contexts, methods and frameworks from choreography and dance can enable a repositioning of engineering approaches to help draw attention back to problem identification and the assumptions guiding this. They argue that such approaches are necessary to improve sustainability challenges: 'engineering needs to change its assumptions about the kinds of effects it can have on urban environments . . .it will require transformative epistemological and methodological approaches that allow new options to emerge from design processes.' (2019, 201).

We align with these approaches of balancing the object and subject-oriented approach in design education. We suggest that to examine somatic or body-based methods in the way they affect designers' ways of thinking requires stepping out of the dominant design epistemology which focuses on how to solve problems – not how to identify problems. Our approach was to work with body-based methods – methods of mapping and designing which foreground working with our own bodies as tools – to address this question of balancing process-orientation in design and teaching design capabilities, with the dominant disciplinary and technical-oriented approach of problem solving. Our research proposes that as design educators we can and should balance the problem-solving possibilities and the problem-identifying possibilities of design methods. In our case, the application of body-based methods in teaching was a springboard to exploring how this balancing act can facilitate a meta-level of critical reflection in design students.

We believe that this meta-level of reflection can help students to ask more questions and know their limitations (and the profession's limitations) thereby putting focus on the process (and learning) rather than solely the object/end design. Therefore, we ask the following research questions: What can we learn from the implementation of body-based methods within design education? How far can body-based methods enable a meta-critical reflection during the design process?

In PBL, we find an opportunity to introduce such ways of thinking. Although students have access to a range of methods to identify problems from real world situations, they are still oriented towards the outcome – the solution – and have limited access to complementary techniques and approaches that can help them reflect on how and why they have identified the problem from their research. Our paper contributes to advance and expand PBL approaches within architectural and urban design education.

2 BODY-BASED METHODS IN THE TEACHING CONTEXT

The implementation of body-based methods was introduced as part of a design course module called "Shaping Urban Public Space II: Conceptual Design and Relations between Buildings and Urban Public Space" (own translation from Danish). The five-week course module is part of the 5th semester bachelor in architecture and design education of Aalborg University, in Denmark. The course module aims to introduce students to design theories and tools to study the relation between buildings and urban public spaces. The course established a thematic focus: working with different people's and beings' perspectives to design urban public space for diverse bodies and species. In this regard, students were introduced and had to work with selected literature rooted in feminist and queer theories related with biodiversity in the city, design for neurodiverse, sensory and motor impaired users, and excluded groups like women, children and queer and diverse ethnic communities. Working with theories and methods introduced in the course, the students were tasked with redesigning an outdoor urban space (a car park). The project submission material included a physical model and working models, one-page design with a conceptual image, one-page analysis and one-page reflection about theories and methodologies used.

Body-based methods were introduced through lectures followed by hands-on exercises and joint discussions. A range of contextual examples from spatial design and from dance scholarship literature were presented, divided in three areas: 1) the moving body as a tool for mapping and measuring using scores and choreography (Hunter 2019; Sara and Sara 2015; Tang 2021); 2) the moving body as a way to help to document and record experience of space using walking, photography from eye level or by using other devices, haptic mapping and scoring (De Wit 2023; Jenkins 2018); and 3) the moving body as a design intervention in its own right which changes the experience of a space (or space-event) using speculative choreography and audio recordings (Chambers & Gerecke 2018; Biggs 2015; Sara and Sara 2015).

Based on the examples, the students were given a series of tasks requiring them to use their own moving body and their subjective experience of moving in space as a way of analysing the site. One task invited students to observe movement in space, developing their own observation methods from a static and from a moving position. This introduced the idea of perceiving space differently whether one is moving or not moving, but in both cases the students were applying themselves as 'objective' observers of their surroundings, with the goal of 'capturing' factual information around them into a retrievable form. The requirement to document movement while moving provided an opportunity to recognize the process of documentation as part of the experience of space inherent in any site analysis exercise.

Experimenting further with their moving body as a way to both experience and document space, the students were invited to create their own mapping 'scores' with 'instructions' about how they would move through the space, and what they would pay attention to (with concepts of 'score' drawn from dance scholarship examples). Most students chose a perspective (e.g., wheelchair user, an animal, a child) as an inspiration to guide and explore the creation of the 'score' as a site mapping tool in urban design. The emphasis was on attending to subjective, body-based knowledge. Experiences documented through a 'score' as map would

communicate something about the site, similar to other forms of site mapping (drawing, video) the students already used. The examples and practical experimentation were offered with the intention of creating opportunities for the students to question what assumptions they held about site analysis and site maps. By focusing on their own subjective, body-based experience, they had to ask: how do I know what about this site is relevant? By devising methods of documenting their experience, they had to ask: what is possible to record? By considering how their documentation could form a communication about the site, they had to ask: what is it about the site that I can, and want to, communicate, and why?

All of these questions were not, in fact, answered by the methods and examples offered. In comparison to traditional, more empirical site analysis and site mapping, these body-based methods of 'analysing' and 'mapping' site were flawed and incomplete. Realizing and understanding this could open up spaces to question and reflect on the practices with which students are familiar, to ask about the flaws and incompletenesses of *all* methods that they may use to analyse and map site, especially when considering different and diverse bodies.In this way, the embodied engagement of using the experiential methods is twofold: on the first level, working with body-based and experiential methods of analysing site trained the students' sensorial sensitivity. On the second level, reflecting on the 'gaps' made obvious by these types of site analysis and site mapping methods trained the students' critical-creative self-reflectivity.

3 METHODS

A focus-group interview was conducted with selected students who worked with body-based methods during their project development and design process. Five (5) students were interviewed, 2 female students and 3 male students (participation was voluntary). Students were interviewed online. The interview was conducted in English and recorded with the consent of the students. A semi-structured interview guide was used for the interview. The interview guide was divided in three themes: 1) reasons and motivations for using body-based methods, 2) challenges encountered during the implementation of methos, and 3) reflections for future use of the methods for design. The small sample size of five students (10 were invited to participate) was due to the nature of this research project, which was explorative and based on an existing teaching module in which students were not required to work with body-based methods.

The analysis of the interview transcripts was done using a thematic-coding approach (see e.g. Gibbs 2007), in which the interview transcripts were analysed for common or repeating themes across responses. The open-ended questions led to a wide range of topics covered in the discussion. The analysis was conducted with the aim of formulating/generating a hypothesis and a research question to work with in future research. Our focus was on understanding how individual students responded to the task rather than extrapolating generalisable facts and claims. We were interested in understanding what happens when new interdisciplinary approaches are used during the design process. Participants' names have been anonymized in the analysis.

4 ANALYSIS AND FINDINGS

The discussion brought about an insight into the ways in which the students thought about their own role as a designer, the existing assumptions they had, and the kinds of questions that the tasks of this course had facilitated them to ask (or not). A core theme that arose was Translation: several students worked with body-based methods to attempt to 'translate' the sensations and experiences of other individuals (such as disabled people or children) to their own experience. Hanna explained that having temporarily removed a specific sensory ability (e.g. hearing, sight), that they had the experience of having that particular sense disability.

Hanna assumed that by amending their own physical experience from their 'norm', they would be able to gain objective knowledge about another's physical experience. This in turn demonstrates the underlying understanding of their role as a designer, which is to find out the objective

(that is, universally experienced, factual) qualities (or affordances) of a site, as it makes itself available to a particular kind of being. This is different from seeing the affordances of a site as inherently subjective, in which case the student would recognize that temporarily removing a sense they in fact usually have would amend their own experience and therefore contribute to an understanding of site as it makes itself available to *themselves*, rather than giving them objective information about the site as it makes itself available to a sensory impaired person.

However, the same student also acknowledged that just by removing that sense stimuli to experience the site, they "could not be sufficiently disabled" meaning that because they haven't had a long-life experience of sensory impairment they did not possess certain skills that those with a sensory impairment would have, for example in using an aid (e.g. a cane or a hearing aid). To overcome this gap, their solution was to interview disabled people about their experience (one of the recommended approaches in their design education). The student had learned that to create a 'successful' design they needed to get as close as possible to the objective experience of the individual for whom they were designing, albeit using different methods.

Other students showed awareness of the impossibility of a 1:1 translation of experience. Kim recognized that the method they used to try and understand the experience of someone using a wheelchair was flawed. Similar to Hanna, the solution Kim identified to this flaw, was to use a real wheelchair instead of the office chair that they had at their disposal. For both Hanna and Kim, the solution to the problem of the limitations of the method was assigned to the apparatus, rather than the method itself. With an understanding of this knowledge 'gap', Hanna considered interviews as an alternative and more successful method. The methods that Hanna and Kim were familiar with from their education did not facilitate them to make the next critical step of asking about the values that underlay the perceived need to understand as closely and objectively as possible the experience of the 'other'. They also did not ask how they as designers could decide what would constitute a 'good' spatial experience for others on the basis of this data collected – how they could decide what problem they would be 'solving'.

For some students, the question about translation did begin to open up the possibility of questioning what the purpose might be for a designer to try to 'translate' experiences in the first place. In investigating the non-human perspective within their design, Sol explored a method of mimicking an animal's behavior, which quickly led them to understand experientially that they would never be able to get into the mindset of that animal. However, the *attempt* to do so, or thinking about it, had a different kind of use, as it made them think more carefully about how an animal and a human differ. Sol did not try to 'solve' the problem of not being able to have the same experience as the animal, but rather was able to use the recognition of the *impossibility of translation* to reflect on what is human and what isn't human about certain experiences of their space.

This line of questioning laid the pathway towards asking about how a designer can generate understanding about different experiences of a particular space or spatial intervention, and why this would be needed for a design as a practice. Sol was prompted to ask about the ethics of whether a human designer should in fact try to design for an animal, when it is not possible for them to fully leave behind their anthropocentric perspective. The experience of becoming more aware of what we can't know, to better understand that a designer can never really know or control how their design works in the world, was "humbling", an observation which was a starting point for a consideration about the ethos and values inherent in the designer's role and practices.

A further topic noticeable in our analysis of the group discussion was a growing but still limited awareness of the interplay between objectivity and subjectivity. While the students were aware of the role of their own subjective experience using some of the body-based and other experiential methods and embraced this role in its ability to give them new ideas about their designs, the dominance of objectivity as framing value continued to be evident. Elian expressed that working with body-based methods made them more aware of their own "subjective bias" in collecting apparently objective data. While the body-based methods had increased awareness and self-reflection on their own subjectivity, the students' own analysis still prioritized objectivity as the dominant form of analyzing and mapping site.

In general, it was evident from the discussion with the students that certain questions or assumptions about what the designers were attempting to do, and the best ways of doing this,

were beginning to be opened up for the students through the course, their assignment, and the questions we were asking them. They had been able to take on board the idea of using their own subjective and body-based experience as data towards their designs. They were able to acknowledge both the possibilities that such methods offered them towards understanding site and proposing creative ideas for their designs, as well as the limitations of the methods they used, especially with regard to being able to fully understand experiences of other individuals whose experience of urban space is routinely different from the students' own.

However, the conditions for reaching a meta-critical approach, e.g. to reflect on wider implications of the limitations of their methods on the role and the practice of designers, were more limited (this was partly a factor of the short time frame of the module). Generally, the students reflected on how they as individual designers could identify and solve particular problems (such as how a sensory impaired person could find their way in a public space). They did not tend to reflect on the ways in which problems had been identified, why these problems were important for designers to solve or indeed how they might be identified and solved differently. This indicates to us some of the inherent limitations of the PBL method. Nevertheless, it became clear through the discussion that due to an openness towards the more subjective and less qualitative and quantitative nature of the site research that they undertook in their designs, that the students were able to begin bringing a certain critical attitude to the way in which they chose and carried out the creative methods, and the relationship between the methods they used and the design that they came up with.

5 DISCUSSION

Through our analysis of the students' responses, we became aware that there are certain values underpinning the critical reflection of the students on their role and on their practices as designers. These values played a strong part, for example, in how the students chose to take perspectives of (relatively disadvantaged) 'others', and having taken those perspectives, in how they conceived of what it meant to successfully make a design responding to the experiences of such 'others'. These values, although unspoken, are closely linked to the assumptions of what design is for. Such values and assumptions are underpinned by the PBL method that encourages students to think in object-oriented ways, rather than to 'stay' with process and develop their critical thinking skills.

However, the interviews also showed that through adopting body-based methods of investigating site, students could begin to recognise and question some of these values. In our case study, we have found that introducing methods of body-based site mapping primarily operated to create the possibility for new ways of thinking about design on a meta-critical level: what leads us to identifying the problems that we are working to solve, and what other possibilities are there for both problems and solutions that we can work on? We propose that by providing the students with tools that are 'flawed' for the purposes that they assume tools in design are supposed to be good for, we give them the conditions for pausing and reflecting on a meta-critical level.

Initially, the students saw the body-based methods as mapping tools for doing site analysis, to complement others such as quantitative or qualitative methods. In this view, adding in body-based methods can 'round out' the designer's toolbox, giving her a better chance of 'fully' understanding or mapping her site. This in turn implies that the designer has a better chance of creating a 'successful' design for that site. This way of thinking relies on a knowledge regime made up of many different assumptions which support the object-oriented mode of thinking about design, in which design is regarded as an intervention into the real world to improve it (see e.g. Bingham-Hall & Cosgrave 2019, Savin-Baden 2020, and Avermaete 2021). In order to decide on how it needs improving, the designer first needs to gather (objective) information about the world. Through our small case study working with students, we are able to show the kinds of thinking processes that can happen if we deviate from the outset from such knowledge regimes.

Firstly: body-based methods for analysing site are already 'flawed' within the above outlined knowledge regime, because they cannot give access to an inherent (objective) truth about a site, only

about a contextual and conditional set of 'truths' related to who is experiencing, when, and how. On realising this, for several of the students the first approach was to think about how the tool can be modified, adapted or complemented to make it 'less flawed', and to come closer to 'accuracy'. This is what we are thinking of as a 'level one' approach to criticality: the students recognise that the tool is limited and move to reduce such limitations in their tools. This approach sits firmly within the conventional knowledge regime for design, as mentioned above.

We are interested in creating the conditions for a 'level two' approach. In this approach, the realisation that the tool (the body-based methods) is 'flawed' leads *also* to a thought process that asks why and how the tool is supposed to be useful in the design process. Once the students could question what the tool was *supposed* to be doing in the context of site analysis – which was triggered by the 'flaws' of the tool being immediately obvious – some were able to access a level of reflective criticality that enabled them to think outside of the knowledge regime outlined above. Sol realised early on in their process that the methods they employed to gain an 'accurate' picture of the experience of non-human beings in the site would fail, since a human cannot access the experience of an animal. Continuing with the design process nonetheless, they expressed a simultaneous level of critical reflection on the function of the design, and therefore also about the knowledge regimes that support an interventionist approach.

Such a reflective capacity is what opens up, eventually, the possibility for reflecting on and finding new ways of looking at the functionalist approach of design that seeks to solve problems. Sol's line of critical reflection, stimulated by the inherent limitations of the methods they were using to analyse their site, offers the questions: what is this design for? What problems am I solving? Are these the 'right' problems to solve? What other problems might there be to solve?

Additionally, with their now growing awareness of the ethical dimension of the work of the designer in this context, the 'level two' critical thinking offers the conditions to ask: Can I ever know what are the 'right' problems to solve? Are there any 'right' problems to solve? What is the role of a designer if we cannot know the truth of a site, or if there is no 'truth' of a site in the first place?

6 CONCLUSIONS

In the examples of the students' thought processes we have described, we noticed that it was important to maintain a balance between an interventionist approach (object-oriented) – which assumes that the design must be functional, solving a problem (namely in bringing about an improvement to a site) – and a *speculative* approach, which allows the design process to continue towards an object, whilst simultaneously keeping space open for reflecting critically on the 'what-ifs' of the problem identification within dynamic and complex situations.

Therefore, we propose that it is important to create balanced conditions for both intervention and speculation within design education. Design itself then becomes a way of discovering, asking new questions, reflecting not just on the design that is underway for the site at hand, but simultaneously about the knowledge regimes that govern what design is as a practice in the world. We find that it is extremely important to ensure that design students are equipped with this capability in order that they can ask questions at 'level two'. We propose that in this way, design as a practice can become more attentive and therefore more responsive to the dynamism of the future changing world. Awareness of the limitations of the design profession can also be developed in this way, allowing future designers to understand that they cannot solve complex problems alone, and encouraging them to more fully understand the responsibilities and power that designers hold in identifying problems.

It is important to note the limitations of this exploration. Firstly, the course module was very short in time. Students had only 4 weeks to explore the methods and develop a design outcome according to the course's requirements and learning goals. In addition, during the process of observing and assisting students in the implementation of body-based methods, we realized that having more time for training, preparation and facilitation of thinking processes was very important to help them develop the level of reflection and criticality.

To unlock the potential of body-based methods and deal with the limitations of dominant design epistemologies, we argue that reflective practices using body-based methods should be

embedded within existing design activities. In particular, creating conditions for reflection requires teachers' facilitation not only to assist and prompt students in the thinking process, but also to learn from it themselves and incorporate these different ways of thinking and methods in further design education activities. In our next iteration of the course taking place next year, we will put emphasis on preparing students to use the body-based methods, and will facilitate the reflection at the meta-critical level further by providing a template with open questions, drawing on those outlined in the discussion section, with suggested points at which students should pause their focus on the *object* of the design outcome, and instead spend time reflecting on the *process*, and on the assumptions and values that contributed to the identification of their problem, and led them towards certain solutions.

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